The σ-dial at the Hou–Luo corner: a bench note on the 2D Boussinesq equations with fractional thermal diffusion between walls

Khalid Ahmed
Independent Researcher
k@shipleed.com
ORCID iD: 0009-0005-3291-7993

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Abstract

We report a certified numerical bench for the two-dimensional inviscid Boussinesq equations in the Hou–Luo boundary-blowup configuration — a periodic strip with no-flow walls and the Luo–Hou driver data — when the temperature equation carries a fractional thermal diffusion $-\kappa(-\Delta)^{\sigma/2}\theta$ with no-flux walls and $\sigma \in [0, 2]$ a dial. The level is certified by a registered battery of machine-floor anchors at every σ, including the degenerate $\sigma = 0$ operator. On it we record four things. First, on the saturated branch the certified driver-collapse observables depend on (κ, σ) only through $X = \kappa k_d^\sigma$, k_d the driver wavenumber, so that the response law $\Omega_\infty\kappa = 66.3\, k_d^{2-\sigma}$ holds down the dial, confirmed at $\sigma = 0.5$ to 0.1% over a twofold range of κ. Second, the structure of the solution does not collapse: at matched physical time and matched physical band the x-spectra of the vorticity and of the wall temperature are ordered strictly monotonically in σ — the larger σ, the steeper the spectrum — at every mode above the amplitude floor and at every certified time, the vorticity ordering being converged between 512^2 and 1024^2 grids to 0.15% mode by mode; its magnitude grows backward in time on a fixed set of modes, and at a fixed time it decreases weakly with the driver damping. Third, a "valley" in a fitted analyticity-strip parameter reported in earlier drafts of this program was a fit-degeneracy artifact and is withdrawn; we record the lesson as a methods section. Fourth, the wall-temperature spectrum at 512^2 is over-steepened for $\sigma \geq 1.375$ by a resolution effect largest at the wall row and decaying through the temperature's content layer, whose anisotropic signature we measure: it is carried mostly by the along-wall resolution, least by the wall-normal resolution of the layer, and in part by both together. The vorticity spectrum, whose layer is grid-converged in each direction separately, is the clean carrier. No analyticity-strip width and no functional form for the σ-separation's time or X dependence are claimed.

1 Configuration and the σ-dial level

The bench. The 2D incompressible Boussinesq equations on the periodic-in-x strip, $x \in [0, L_x)$ with $L_x = 1/6$ (the Luo–Hou z-period [9]), $y \in [0, 1]$ between two walls:

$$\omega_t + u \cdot \nabla\omega = \theta_x, \qquad \theta_t + u \cdot \nabla\theta = -\kappa(-\Delta)^{\sigma/2}\theta,$$
$$u = (-\psi_y, \psi_x), \qquad \Delta\psi = \omega, \qquad \psi = 0 \text{ at } y = 0, 1 \text{ (no-flow walls)},$$

with no-flux walls for θ, ω inviscid, and the exact Hou–Luo driver data [9]

$$\theta^0 = A\, e^{-60(y(2-y))^4} \sin^2(2\pi x/L_x), \qquad \omega^0 = 0, \qquad A = 10^4;$$

the symmetry class is ω odd and θ even in x. The corner is $(x, y) = (0, 0)$; the driver wavenumber is $k_d = 4\pi/L_x = 24\pi = 75.398$. Discretisation: Fourier in x, Chebyshev–Gauss–Lobatto in y

($N+1$ points), 2/3 dealiasing in both directions, fourth-order Runge–Kutta with an adaptive CFL step (cfl $= 0.4$ throughout), the Poisson solve by a Chebyshev-tau quasi-inverse certified against a dense-LU reference. The base solver's anchor battery (exact solutions, invariants, manufactured solutions, resolution floors) is our earlier record and is not repeated here.

The certified window and the verdicts. Every run carries its own certification: a diagnostic record is *certified* while the spectral tails in x and y are both $\leq 10^{-6}$ of the field; t_{cert} is the last certified time, $\Omega_{\text{cert,pk}}$ the certified peak of $\sup|\omega|$, g_{cert} the certified growth factor, itx $= \int \sup|\theta_x|\,dt$ the driver impulse. Verdicts at the stop: SATURATED (the driver's impulse and $\sup|\omega|$ both stop growing to within 2% over the trailing 25% window — the κ-world's analogue of decay, since ω has no dissipation channel), RESOLUTION (the tails pass 10^{-2} — the front has reached the grid while still growing), BLOWUP-bar and UNDET($T_{\max}$) (never fired on this program). Only certified quantities enter any statement below; the uncertified endgame after t_{cert} is used for one explicitly labelled observation (Section 6). Throughout, a *registered letter* is an outcome label whose deciding statistic and bar were fixed before the comparison was run; we report every letter as it fired.

The σ-dial level. The fractional operator is defined spectrally on the strip: with the two Neumann rows eliminated, the reduced y-operator is $A_{\text{red}}(k_x) = A_0 + k_x^2 I$ — the eigenvectors are k_x-independent, A_0's spectrum is real and non-negative ($\max|\operatorname{Im}\lambda| = 0$ exactly), it matches the Neumann eigenvalues $(n\pi)^2$ to $2.5 \cdot 10^{-13}$ / $3.3 \cdot 10^{-13}$ / $1.5 \cdot 10^{-12}$ at $N_y = 32$ / 64 / 128, and the eigenbasis is well conditioned ($\operatorname{cond} W = 2.66$ / 3.36 / 4.18). The diffusive half-step of the Strang split is therefore the exact exponential $\theta \leftarrow W \operatorname{diag}(\exp(-\kappa(\lambda_n + k_x^2)^{\sigma/2} h))W^{-1}\theta$, unconditionally stable at every σ including $\sigma = 0$. Convention, stated because it is a choice: $0^{\sigma/2} := 0$ for every $\sigma \geq 0$, so constants are annihilated at every σ and the $\sigma = 0$ operator is $I - P_0$. $\sigma = 2$ is the ordinary Laplacian; in the $|D|^\beta$ convention of the well-posedness literature our σ is β (Section 7).

Certification (the registered battery). All items met at their predicted floors, three after an honest test redesign recorded in the run log: (a) the eigen-decomposition as above; (b) pure-eigenmode decay against $\exp(-\kappa\lambda^{\sigma/2}t)$ at $\sigma \in \{0, 0.5, 1, 1.5, 2\}$: $4.57 \cdot 10^{-14}$ worst case; (c) the $\sigma = 2$ level against the certified Crank–Nicolson κ-level: dt-halving ratios 3.99 / 4.00 / 4.00; (d) $\kappa = 0$ bit-identical to the inviscid solver (0.0); (e) the $\sigma = 0$ degenerate operator on random compatible data $1.14 \cdot 10^{-14}$ / $7.25 \cdot 10^{-15}$ with the operator identity $|D(h)v - (P_0 + e^{-\kappa h}(I - P_0))v| = 2.49 \cdot 10^{-14}$; (f) no-flux enforcement from wall-incompatible data $7.54 \cdot 10^{-15}$; (h) the eigenvalue error at $n = N_y/3$ falling $3.06 \cdot 10^{-9} \to 6.9 \cdot 10^{-15} \to 6.3 \cdot 10^{-16}$ under doubling. One registered item was REFUTED and converted: (g), exact conservation of the θ-mean, fails at the $3 \cdot 10^{-11}$ ($\sigma = 0$) to $1.2 \cdot 10^{-8}$ ($\sigma = 2$) level on compatible data because the strip-integral functional is not a left null vector of A_0; the mean drift is carried in every run as a free resolution monitor, and it is σ-restricted: under a grid doubling it falls at $\sigma = 2$ and rises at $\sigma = 0.5$ (the weakly damped spurious modes), so the monitor is read only at $\sigma \geq 1$.

2 The saturated-branch collapse and the σ-extended saturation law

The κ-dial law at $\sigma = 2$. On the saturated branch the certified plateau obeys $\Omega_\infty\kappa = 66.25$ / 66.4 / 66.4 / 66.3 at $\kappa = 0.25$ / 0.3 / 0.4 / 0.6 (256^2, $A = 10^4$), constant to 0.3% over $\times 2.4$ in κ, with the exact amplitude symmetry $\Omega_\infty\kappa = c_0 A$, $c_0 = 6.63(2) \cdot 10^{-3}$, confirmed at $A = 2500$ to 1.0–1.1%. The law constant's certified N-progression at $\kappa = 0.1$ is 88.6% $\to$ 91.7% $\to$ 93.6% of the predicted plateau at 512^2 / 1024^2 / 2048^2 (the window physics; the uncertified continuation touches 100–102%). A measured near-tightness of the vorticity bound $\frac{d}{dt}\sup|\omega| \le \sup|\theta_x|$ (0.94–0.96) and the driver form itx $\cdot\, \kappa \approx 7.0 \cdot 10^{-3} A$ are the mechanism candidates on the record; no mechanism is claimed here.

The collapse. At the reference setup with $N = 256$, the certified observables $\Omega_{\mathrm{cert,pk}}$, t_{cert}, g_{cert} depend on (κ, σ) essentially only through $X = \kappa k_d^{\sigma}$: at $X = 1421.223$ the five rows $\sigma = 2$ / 1.5 / 1 / 0.75 / 0.5 (a 655× range of κ) all SATURATE with $\Omega_{\mathrm{cert,pk}} = 265.03$ / 265.54 / 265.63 / 265.38 / 265.57 (spread 0.23%); at $X = 1136.978$ they give 328.19 / 328.82 / 329.19 / 329.19 / 330.16 (spread 0.60%). The collapse is domain-bounded to the saturated branch: at three deeper $X = 3410.9$ / 2274.0 / 1705.5 the $\sigma = 0.5$ vs $\sigma = 2$ deviation is -0.127% / -0.151% / -0.045%, but on the resolution branch it is $+1.322\%$ at $X = 852.7$ and $+0.471\%$ at $X = 568.5$, and under a doubling to $N = 512$ the $X = 568.5$ pair degrades to $+3.785\%$ while the $X = 1705.5$ pair improves to -0.023%. The reading originally drawn from the collapse ("the controlling wavenumber is the driver's, not the front's") was withdrawn by a registered control: on the saturated branch 99.0–99.99% of the wall temperature's x-energy is still in the single driver mode, a branch that cannot distinguish the front from the driver. Two grid-convergence points of the collapse are added in this note (Section 5): the seven-row $X = 1421.223$ column at $N = 512$ has $\Omega_{\mathrm{cert,pk}} = 265.51$–$265.83$ (spread 0.12%), +0.11% over the five $N = 256$ rows' mean, and the seven-row $X = 1705.468$ column at $N = 512$ has $\Omega_{\mathrm{cert,pk}} = 221.33$–$221.65$ (spread 0.15%), +0.14% over the two $N = 512$ rows ($\sigma = 0.5$ / 2) banked at that X.

The σ-extended saturation law. Composing the κ-dial law with the collapse gives $\Omega_\infty \kappa = 66.3\, k_d^{2-\sigma}$. Registered before the runs at $\sigma = 0.5$ (predicted $66.3\, k_d^{1.5} = 4.3407 \cdot 10^4$, bar 3%), the measured plateaus are $\Omega_{\mathrm{cert,pk}} \cdot \kappa = 4.33869 \cdot 10^4$ / $4.34170 \cdot 10^4$ / $4.34478 \cdot 10^4$ at $\kappa = 392.8194$ / 261.8796 / 196.4097 — deviations -0.045% / $+0.024\%$ / $+0.095\%$, mutually consistent to 0.140% over a 2.00× range of κ; the $\sigma = 2$ reproduction obligation was discharged first (66.354 / 66.416 / 66.393 at $\kappa = 0.60$ / 0.40 / 0.30, i.e. +0.08 / +0.18 / +0.14% against 66.3). **Consequence, stated exactly:** the inverse-first-power form holds at a σ where it had never been tested, so on the accessible saturated branch there is no finite-κ divergence at $\sigma = 0.5$ either; $\kappa_c(\sigma) = 0$ down the dial is an extrapolation of a confirmed law, never a measurement, and it is labelled so wherever it appears.

3 The structure that does not collapse: the σ-ordering of the spectra

The object. For a row (κ, σ) and a field f, let $\mathrm{amp}_f(k)$ be the amplitude of the x-Fourier mode k: for θ the modulus of the transform of the driver-wall row (the wall row carries the θ structure, Section 6), for ω the maximum over y of $|\mathcal{F}_x \omega|$ (the solver's own strip object). The driver-normalised log-amplitude is $a_\sigma(k) := \ln(\mathrm{amp}_\sigma(k)/\mathrm{amp}_\sigma(k_d))$; the matched physical band is $B^* :=$ the even modes 52..102 (26 modes; the upper half of the 512^2 fit band, used at every N); modes below 10^{-12} of the driver mode are masked, and a letter is defined only when at least half of B^* is above the mask for every compared pair (the usable fraction is printed with every number). The **ordering statement** is the sign of $S(k; \sigma, \sigma') := a_\sigma(k) - a_{\sigma'}(k)$ at every k of B^* for adjacent σ; its **magnitude** is the band mean $\bar{S}$. Every comparison is made at a matched time — $t_m :=$ the minimum t_{cert} over the compared rows, every row's spectrum log-linearly interpolated between its two bracketing diagnostic records, both records also evaluated (the timing spread is printed) — and, across grids or columns, at the same physical modes. One-shape fits ($\ln \mathrm{amp} = a - \delta k$ on B^*, its upper quarter, or the row's own band) are reported beside the model-free object with their formal σ_δ; no fitted slope is read as an analyticity-strip width (Section 4).

The ordering at matched time. Columns in σ at fixed $X = \kappa k_d^\sigma$, $N = 512$, $A = 10^4$, cfl $= 0.4$, the certified stash of every run's diagnostics:

- $X_2 = 0.10\, k_d^2 = 568.49$ (seven rows $\sigma = 0.875$, 0.9375, 1, 1.125, 1.25, 1.375, 1.5; all RESOLUTION; $t_m = 0.003714$): adjacent-pair $\bar{S}$ on the θ wall -0.6621 / -0.6147 / -1.1152 / -1.0165 / -0.9815 / -1.0198, on ω -0.1222 / -0.1159 / -0.2151 / -0.1958 / -0.1790 / -0.1643; every mode of B^* negative at all three time samples, usable 1.00. Column separation $\bar{S}(1.5, 0.875)$: θ -5.41, ω -0.99. **Both MONOTONE.** One-shape slopes on B^*: θ $0.11591 \to 0.14250$ (+22.9%; σ_δ 0.0008–0.0010, R^2 0.9983–0.9992; the upper-quarter band $0.10744 \to 0.13557$, consistent to 7%), ω $0.07994 \to 0.09194$ (+15.0%; ± 0.00023), the solver's own-band ω strip slope $1.9761 \cdot 10^{-3} \to 2.2931 \cdot 10^{-3}$ (+16.0%).
- $X_3 = 0.15\, k_d^2 = 852.73$ (six rows $0.875 \ldots 1.5$; all RESOLUTION; $t_m = 0.004488$): θ -0.9938 / -0.9220 / -0.9211 / -1.0082 / -1.2239, ω -0.1631 / -0.1467 / -0.1326 / -0.1202 / -0.1094; column separation θ -5.07, ω -0.67. **Both MONOTONE.** Slopes θ $0.11254 \to 0.15526$ (+38.0%), ω $0.07911 \to 0.08762$ (+10.8%), strip slope $1.9568 \cdot 10^{-3} \to 2.1810 \cdot 10^{-3}$ (+11.5%).
- $X_4 = 0.20\, k_d^2 = 1136.98$ (seven rows $0.75 \ldots 1.5$; SATURATED $\times 5$, RESOLUTION $\times 2$ — mixed stop criteria, so report-grade; $t_m = 0.005078$): θ -0.9453 / -0.8765 / -0.8844 / -0.9835 / -1.1768 / -1.4281 (usable $1.00 \ldots 0.73$), ω $-0.1448 \to -0.0854$ (usable 1.00). **Both MONOTONE.** Slopes θ $0.12381 \to 0.19622$ (+58%), ω $0.09327 \to 0.10197$ (+9.3%).
- $X_5 = 0.25\, k_d^2 = 1421.22$ (seven rows $0.75 \ldots 1.5$; SATURATED $\times 7$ with $t_{\mathrm{cert}} = 0.005046$–$0.005047$, the stop — a letter at this matched time is a statement at the rows' saturation stop; $t_m = 0.005046$): ω -0.1476 / -0.1314 / -0.1177 / -0.1059 / -0.0957 / -0.0866 (usable 1.00), column separation $\bar{S}(1.5, 0.75) = -0.6849$ — **MONOTONE**; the θ wall at this late matched time has under half of B^* above the mask for the two top pairs (usable 0.38 / 0.27) — **UNDEFINED(floor)**, no θ letter. Slope ω $0.14888 \to 0.15776$ (+6.0%; ± 0.00022).
- $X_6 = 0.30\, k_d^2 = 1705.47$ (seven rows $0.75 \ldots 1.5$; SATURATED $\times 7$ with $t_{\mathrm{cert}} = 0.005047$–$0.005048$, the stop; $t_m = 0.005047$; the $\sigma = 2$ row banked at this X was re-run through the diagnostic stash first and reproduced every saved key bit for bit): ω -0.1407 / -0.1253 / -0.1122 / -0.1010 / -0.0911 / -0.0825 (usable 1.00), column separation $\bar{S}(1.5, 0.75) = -0.6529$ — **MONOTONE**; the θ wall at this matched time has under half of B^* above the mask for every pair (usable $0.38 \ldots 0.04$) — **UNDEFINED(floor)**, no θ letter. Slope ω $0.20284 \to 0.21132$ (+4.2%; ± 0.00022).

At the earlier time $0.8\, t_m$ the ω ordering is unchanged at X_2, X_3, X_4 (MONOTONE; the separations 1.4–1.6× larger) while the θ wall drops below the amplitude floor for the top pairs (UNDEFINED(floor) — a floor statement, not a reversal).

Reading, exact. At one time inside every row's certified window and at the same physical modes, the x-spectra of the vorticity (at all five driver dampings we ran) and of the wall temperature (at the three where it clears the floor at the matched time) are ordered strictly monotonically in σ: the larger σ, the steeper the spectrum (the lower the amplitude at fixed k relative to the driver). This is the "structure separation" of our earlier record in its ordering form, with the three receipts our instrument rule demands: formal uncertainties on every fitted number and the model-free object's k-scatter, model robustness (the model-free object, two one-shape bands and the solver's own band all monotone), and matched stop criteria and times. Its magnitude is stated only as the matched-(t, k) numbers above.

***N*-convergence at matched time and matched band.** The X_3 pair $\sigma \in \{0.875, 1.375\}$ re-run at $N = 1024$ (two rows, both RESOLUTION) and compared at $t_m = 0.004488$ on B^*: the ω spectrum is FIELD-CONVERGED mode by mode — $D_N := \max_{B^*} |a_{1024} - a_{512}| = 0.0013$

($\sigma = 0.875$) and 0.0015 (1.375), i.e. 0.13–0.15% — and its σ-separation ratio $r_N := \bar{S}_{512}/\bar{S}_{1024} =$ 1.000 (-0.5626 / -0.5627; 0.999–1.000 across $\pm 90\,\mu$s of like-for-like timing shifts). The θ wall spectrum is converged at $\sigma = 0.875$ ($D_N = 0.0174$, 1.7%) but NOT at $\sigma = 1.375$ ($D_N = 0.5726$: the 512^2 wall spectrum is steeper than the 1024^2 one by a k-linear excess, $+0.138$ at $k = 52$ to $+0.573$ at $k = 102$ in $\ln \mathrm{amp}$ — a factor 1.77 at $k = 102$ — fitted at $+0.00872 \pm 0.00014$ per mode, R^2 0.994, and growing in time from 0.21 at $t = 0.0030$ to 0.57 at t_m); its separation ratio is $r_N = 1.092$ (-3.8450 / -3.5202). The solver's own-band ω strip slope, an N-scaled band, agrees between grids to 1.052 / 1.040 at the two σ, the residual being the band term of the ω spectrum's weak prefactor. On the 1024^2 rows' own upper band (modes 104..204) the one-shape slopes are lower than on B^* (θ 0.08906 / 0.11249 against 0.11228 / 0.12903; ω 0.07295 / 0.08002 against 0.07912 / 0.08624): the local slope falls with k on both fields, so no band on the record is in an asymptotic exponential regime, at either N. These rows are RESOLUTION-class. The SATURATED-class side was tested on the X_5 pair $\sigma \in \{0.875, 1.5\}$ re-run at 1024^2 (both SATURATED at $t_{\mathrm{cert}} = 0.005012$ against 0.005046–0.005047 at 512^2; $\Omega_{\mathrm{cert,pk}}$ 265.64 / 265.68 against 265.66 / 265.71) and compared at the fixed time $t_c = 0.003714$ on the fixed 16-mode set 52..82: the two ω spectra move together between the grids ($D_N = 0.0089$ / 0.0084, i.e. 0.8–0.9% mode by mode, the same sign at every mode) and their σ-separation is N-converged — $\bar{S}_{512} = -0.7346$ against $\bar{S}_{1024} = -0.7351$, $r_N = 0.9992$ (0.9986 at both common shifts of $\pm 7.2 \cdot 10^{-5}$; registered letter N-CONVERGED at a 0.5% bar) — with the fixed-set course on 52..76 over the common window $[0.003461, 0.005012]$ agreeing between the grids to 0.24% at every one of nine samples (ratios 1.680 / 1.683).

4 Methods: what the earlier drafts got wrong, and the rules that replaced it

The withdrawn "valley". Earlier drafts of this program read a two-shape fit $\ln \mathrm{amp} = a + p \ln k - \delta k$ on the θ wall's fit band $[k_d, 0.2N]$ at each run's own stop and reported δ's interior minimum in σ as a structure ("the analyticity strip is narrowest at $\sigma \approx 1.5$ / 1.375 / 1.25 / 1.125 . . . ", moving with damping and with refinement). An audit of the banked records found: the fit's prefactor and exponential slots are correlated at +0.90–0.93 (a property of the basis on that band, independent of the field — $\mathrm{corr}(p, \delta) = +0.908$ at every 512^2 row); the one-grid-step margins carrying every position statement were 0.2–1.6 of the fit's own formal σ_δ; the minimum moves with the fit band; under every one-shape variant (p fixed at the column mean, p fixed at -4, the pure exponential on the upper half / upper quarter / full band) every resolution-class column is monotone increasing in σ with no interior minimum; the solver's own ω strip series, read at the matched time, is monotone at every column; and at the fourth X the "minimum" sat exactly where the stop criterion switched from SATURATED to RESOLUTION. The rebuild then showed, with the stop-time confound removed, that the two-shape δ-slot still puts its minima at the old positions at matched time (θ at X_2: 0.08981 / 0.08963 / 0.08880 / 0.08660 / 0.08522 / 0.08595 / 0.09031 across $0.875 \to 1.5$, σ_δ 0.0018–0.0025, a minimum at 1.25 with 1.375 within $0.3\,\sigma_\delta$; at X_3 a minimum at 1.125) while the model-free object and every one-shape fit are monotone: **the valley was the degenerate (p, δ) pair's crossing point — a property of the fit, not of the field or the stop time — and every position statement of the earlier drafts is withdrawn.** The lesson is recorded as a rule: a fitted structure parameter enters a statement only with its formal uncertainty printed, a model-robustness check (nested and alternative models, sub-band convergence) and matched stop criteria and times across the compared rows.

Matched band and matched time. Two "×2 between $N = 512$ and $N = 1024$" statements of an earlier record were comparison artifacts: the tail-fit band scaled with N (52..102 at 512^2 but 104..204 at 1024^2; on the same physical modes the ratio is 1.07–1.19), and the strip slopes were read at each column's own t_{cert} (at a common time the ratio is 1.047–1.053). The local slope on a band centred at k_c decomposes as $\delta_{V0} + |p|/k_c$ with the fitted prefactor $p \approx -3.5 \cdots -4.5$ on the θ wall, so every "tail δ" on the record is dominated by $|p|/k_c$ and the asymptotic exponential regime $k \gg |p|/\delta$ (≈ 3000 at 1024^2) lies beyond every N we ran: **no analyticity-strip width has been measured on this bench; what the instruments measure is a local log-slope on a named band at a named time.** Rule: cross-N and cross-column comparisons are made at matched physical band and matched time — cross-column at one physical time — on a model-free object where one exists.

Fixed mode sets across a course, and letters at the instrument's resolution. A band mean taken over a mask that grows in time (the floor mask opening from half of B^* at a window's start) rises before the physics is seen, because $|S(k)|$ grows with k; five of eight registered time-course magnitude letters in an earlier pass fired on that mask edge (recorded as fired; the labelled diagnostic on a fixed set beside them). Rule: a time-course or cross-time letter on a band-mean object holds its mode set fixed across the course; a cross-column or cross-N comparison is made on one fixed set (its size printed as the validity condition) and, for the N-comparison, on one window. Two further rules were added when the cross-column letters were re-read: registered letter clauses are mutually exclusive (one deciding statistic per letter — here the maximum deviation from the mean at a $\pm 10\%$ bar — with the span and the monotonicity reported beside it), and a monotonicity statement across columns is made at the instrument's resolution: every step between adjacent columns is flagged resolved or unresolved by whether the two columns' like-for-like timing ranges are disjoint, all columns being shifted together by one common shift (half the coarsest record cadence among them). Every number of Section 5 obeys these rules.

5 The σ-separation in time and across the driver damping

The ordering is stable in time. On fixed nine-sample grids from each column's floor-window start to its t_m, with every row interpolated at the same grid time and like-for-like timing shifts (all rows together by $\pm$ half a record cadence), the model-free ordering is MONOTONE at every one of the 72 grid samples across X_2 / X_3 / X_4 at 512^2 and the X_3 1024^2 pair, on both fields, and at every one of the nine ω samples of the X_5 and X_6 columns' own fixed-set courses: the σ-ordering holds at every certified time at which half the band is above the floor.

The magnitude grows backward in time on a fixed set of modes. On the common 13-mode set 52..76 the ω column separation $|\bar{S}(\sigma_{\max}, \sigma_{\min})|$ over each column's own certified window [window start, t_m] runs $1.4831 \to 0.8364$ at X_2 ($[0.002369, 0.003714]$; ratio 1.773), $1.1925 \to 0.5613$ at X_3 ($[0.002748, 0.004488]$; 2.125), $1.2141 \to 0.5605$ at X_4 ($[0.003091, 0.005078]$; 2.166), $0.9688 \to 0.5695$ at X_5 ($[0.003461, 0.005046]$; 1.701) and $0.7799 \to 0.5426$ at X_6 ($[0.003844, 0.005047]$; 1.437), non-increasing in t at every sample; the θ wall separation is nearly flat over its shorter windows ($\times 1.05$–1.08 at 512^2). The local log-derivative $d \ln|\bar{S}_\omega|/d \ln t$ steepens toward t_m (from ≈ -1 to $-2.2 \cdots -2.4$ at the 512^2 columns' t_m). The ratios are descriptive: each belongs to its own window, and the windows differ by column (the more damped the column, the later its window opens and the shorter it is), so no cross-column reading of the ratios is made and no functional form in t is fitted.

The time course is N-converged. The X_3 rows (0.875, 1.375) at 512^2 and the same pair at 1024^2 on one window $[0.002675, 0.004488]$ and one fixed 13-mode set: $|\bar{S}(1.375, 0.875)|$ runs $1.0195 \to 0.4704$ (ratio 2.168) at 512^2 and $1.0208 \to 0.4704$ (2.170) at 1024^2, the two courses agreeing to 0.1% at every one of the nine samples (quotient of the ratios 0.999). The larger ratio

$\times 3.60$ recorded earlier for the 1024^2 pair was the pair's longer window (to its own $t_m = 0.005239$), not a grid effect.

The X-dependence at one physical time. The columns compared at one physical time differ in their stop class (X_2 / X_3 RESOLUTION, X_4 mixed, X_5 / X_6 SATURATED); every row is inside its certified window at the comparison time, and the comparison is made on one fixed set of modes across all the compared rows, whose size is printed because the values depend on it ($|S(k)|$ grows with k, so a mean over more modes is larger in magnitude — the 26-, 17-, 16- and 13-mode values below are not interchangeable). At the common time $t_c = 0.003714$ (inside every 512^2 row's certified window) the ω column separation $\bar{S}(1.5, 0.875)$ on all 26 modes of B^* is -0.9924 / -0.9933 / -0.9245 at X_2 / X_3 / X_4 — equal at X_2 and X_3 to 0.1%, 7% lower at X_4; maximum deviation from the mean 4.7% (registered letter X-FLAT at the $\pm 10\%$ bar). At a second common time $t_2 = 0.0033$, on one fixed 17-mode common set (52..84) across the six rows, the values are -1.1151 / -1.0448 / -0.9456 (like-for-like ranges under one common shift of $\pm 5.9 \cdot 10^{-5}$: $[-1.1455, -1.0825]$, $[-1.0671, -1.0208]$, $[-0.9635, -0.9263]$) — within $\pm 8.7\%$ of their mean (9.0% / 8.3% at the shifts; the registered letter X-FLAT at the $\pm 10\%$ bar, as fired), and at the same time strictly decreasing in X with a 16% span, both steps (6.3%, 9.5%) resolved (the registration's X-ORDERED clause was also met; the letter fired by the instrument's precedence, and both facts are stated — the clauses have since been made exclusive, Section 4, under which the same numbers read X-FLAT). The X_5 and X_6 columns lie below their floor windows at t_2 (0.46 / 0.42 of B^* usable at X_5, less at X_6), so the t_2 table has three columns by construction. With the fifth column, at t_c on one fixed 16-mode common set (52..82) across the eight rows: -0.8725 / -0.8688 / -0.8069 / -0.7346 at X_2 / X_3 / X_4 / X_5 — maximum deviation from the mean 10.5% (11.3% / 9.7% under one common shift of $\pm 7.2 \cdot 10^{-5}$: the letter is bar-marginal within its timing resolution), non-increasing in X with the $X_2 \to X_3$ step (0.4%) inside the timing ranges of both columns and the two further steps (7.1%, 9.0%) resolved, span 16.8%: the registered letter X-ORDERED (the labelled expectation X-FLAT did not obtain). With the sixth column: its ω floor window opens at 0.003844, after t_c and after the end of the X_2 column's certified window (0.003714), so no time exists at 512^2 at which X_2 and X_6 can be compared on this bench, and the five-column letter at t_c is UNDEFINED(floor) by its validity condition (the X_6 rows carry 0.46 / 0.42 of B^* above the floor at t_c). At the earliest common time of the four more-damped columns, $t'_c = 0.003844$, on one fixed 13-mode common set (52..76) across the eight rows: -0.7843 / -0.7360 / -0.6732 / -0.6114 at X_3 / X_4 / X_5 / X_6 (ranges under one common shift of $\pm 8.6 \cdot 10^{-5}$: $[-0.8156, -0.7534]$, $[-0.7597, -0.7127]$, $[-0.6915, -0.6544]$, $[-0.6181, -0.5965]$) — maximum deviation 12.8% (14.3% / 12.2% at the shifts), steps 6.2% (unresolved), 8.5% and 9.2% (resolved), span 24.7%: the registered letter X-ORDERED (the labelled expectation). The two four-column tables overlap on X_3 / X_4 / X_5 and are never merged into one row. The product $\bar{S} \cdot X / k_d^2$ varies by 24–38% at every time and set (no $1/X$ law; the near-constancy of that product at each column's own t_m, 6.4% at three columns, is a coincidence of the comparison times — with the fifth column's own t_m, a saturation stop, the spread is 22%). The deficit of the most-damped column is not a floor effect: on stricter common sets (10×, 5× and 3× the mask; 13, 14 and 15 modes) the X_5 deficit at t_c stays at 16%, and every row's spectral tail beyond B^* sits 10^4 times below the mask.

Reading, exact (report-grade; five columns, two overlapping four-column tables at two times; no law). Across the columns we ran, the ω-spectral σ-separation at a fixed physical time decreases weakly with the driver damping: at t_c the two least-damped columns are equal within the timing resolution (0.4%), and the separation is 7.5% lower at 2× and 15.8% lower at 2.5× the damping of X_2; at t'_c on the four more-damped columns it is 6.2% / 14.2% / 22.0% lower at 4/3× / 5/3× / 2× the damping of X_3, the steps from X_4 on resolved. The three-column statement over a 2× range from X_2 reads flat to 5%; the four-column statements over 2.5× (from X_2) and 2× (from X_3) do not at the ±10% bar, the first of them bar-marginal. The earlier record's X-dependence at each column's own matched time (−0.99 / −0.67 / −0.53) was the columns' different matched times, not a property of the separation. We fit no form in X or t; the tables are the statement. The deficit of the saturated columns survives a grid doubling: the X_5 value at t_c is N-converged to 0.08% (Section 3).

What the fifth and sixth columns add to Section 2. All seven X_5 rows at 512^2 SATURATE at the same certified time (0.005046–0.005047) with $\Omega_{\text{cert,pk}} = 265.51$ / 265.66 / 265.77 / 265.83 / 265.83 / 265.79 / 265.71 across $\sigma = 0.75 \ldots 1.5$ — σ-flat to 0.12% and +0.11% over the five $N = 256$ rows at the same X (265.03–265.63): the saturated-branch collapse is grid-converged at this X to 0.1%, its tightest row on the record. All seven X_6 rows at 512^2 SATURATE at 0.005047–0.005048 with $\Omega_{\text{cert,pk}} = 221.33$ / 221.45 / 221.55 / 221.61 / 221.65 / 221.64 / 221.60 across $\sigma = 0.75 \ldots 1.5$ — σ-flat to 0.15% and +0.14% over the two $N = 512$ rows at $\sigma = 0.5$ / 2 banked at the same X (221.22 / 221.27), the $\sigma = 2$ row of which was re-run through the diagnostic stash and reproduced every saved key bit for bit. Read against the law of Section 2, the two columns' means give $\Omega_{\text{cert,pk}} \cdot X/k_d^2 = 66.43$ (X_5) and 66.46 (X_6) against the 256^2-calibrated constant 66.3 (+0.2% / +0.25%), the same sign and size as the +0.1% grid moves above.

6 The wall temperature's resolution layer

The 512^2 over-steepening of the θ wall spectrum at $\sigma \geq 1.375$ (Section 3) is θ-specific and layer-confined. The ω spectrum AT THE WALL is N-converged to 0.13–0.15% at both σ (the wall row carries the max-over-y ω spectrum at every mode of B^*, at both N), so the defect is a property of the θ field's wall row, not of the grid's wall row. The θ x-spectral content on B^* lives in a wall layer thinner than $y = 0.01$ at both N: the max-over-y θ spectrum equals the wall's (D_N 0.5725 against 0.5726 at $\sigma = 1.375$; 0.0174 against 0.0174 at 0.875), while rows evaluated exactly at $y = 0.05$ and $y = 0.01$ carry 0 and 9–14 of the 26 modes above the floor although their amplitudes are 0.985–0.9999 of the wall's. The far wall carries no corner structure (its driver-mode amplitude is 10^{-12}–10^{-11} of the driver wall's). The θ spectral maximum sits on the wall row at every certified time at both N and both σ; an interior maximum ($y \approx 5$–$8 \cdot 10^{-3}$) appears on the 1024^2 $\sigma = 1.375$ field only for $t > 0.0070$, 30% past that row's $t_{\text{cert}} = 0.005375$ — an uncertified-endgame observation, never seen at $\sigma = 0.875$ or at 512^2. Mechanism unclaimed. Bench policy: every θ-wall statement at 512^2 for $\sigma \geq 1.375$ carries this qualifier; the ω spectrum is the carrier of the structure separation.

The layer measured. The four runs were repeated with a y-profile stash — the x-modes $[k_d] + B^*$ of θ and ω evaluated exactly (the Chebyshev representation) on a fixed 166-row physical grid ($y = 0$ and $10^{-6} \ldots 10^{-1}$ log-spaced, 7.5% per step, the same grid at both N) at every diagnostic record — after every banked key of the four rows (the records and the side files) had been reproduced bit for bit through the extended class. At $t_m = 0.004488$, $\sigma = 1.375$, each grid row normalised by its own driver mode and the comparison defined only where at least 13 of the 26 modes clear the 10^{-12} floor at both N: the 512^2–1024^2 difference $D_N(y) = \max_k |a_{1024}(k, y) - a_{512}(k, y)|$ is 0.572 at the wall (the value of Section 3), 0.55 at $y = 10^{-4}$, 0.31 at $2 \cdot 10^{-3}$, 0.28 at $3 \cdot 10^{-3}$ and 0.20 at $5 \cdot 10^{-3}$ (the mean difference +0.33, +0.32, +0.18, +0.15, +0.12), undefined beyond $y_L = 7.5 \cdot 10^{-3}$; the defect's extent at a 0.30 bar is $y_D = 2.1 \cdot 10^{-3}$, $y_D/y_L = 0.27$ — the registered letter MIXED (neither wall-confined, $y_D < 0.1\, y_L$, nor layer-filling, $y_D \geq 0.5\, y_L$; the labelled expectation LAYER did not obtain), the same at one common timing shift of $\pm 4.6 \cdot 10^{-5}$ (0.29 / 0.27 / 0.29) and bar-marginal in y (D_N crosses 0.30 between $2 \cdot 10^{-3}$ and $3 \cdot 10^{-3}$ at every shift; the max statistic carries single-mode spikes where a mode's modulus passes near zero, the mean does not). The control $\sigma = 0.875$: $D_N \leq 0.023$ at every definable row. The layer's thickness — the e-fold depth of the B^* modes from their wall values, the crossing log-interpolated between grid rows, medians over the 26 modes: θ at $\sigma = 1.375$ $5.8 \cdot 10^{-4}$ at 512^2 against $5.1 \cdot 10^{-4}$ at 1024^2 (ratio 1.15 — the coarse grid's θ layer is 15% thicker), θ at $\sigma = 0.875$ $4.9 \cdot 10^{-4}$ / $4.8 \cdot 10^{-4}$ (1.007), ω $9.7 \cdot 10^{-4}$ / $9.7 \cdot 10^{-4}$ ($\sigma = 1.375$) and $1.01 \cdot 10^{-3}$ / $1.01 \cdot 10^{-3}$ ($\sigma = 0.875$), ratio 1.000: the θ content lives within $\approx 5 \cdot 10^{-4}$ of the wall at both σ — half the ω layer — spanned by 7–8 Chebyshev rows at 512^2 and 14–15 at 1024^2; per mode the depth falls with k ($8.0 \cdot 10^{-4}$ at $k = 52$ to $4.8 \cdot 10^{-4}$ at $k = 102$ at 512^2, $\sigma = 1.375$); the median θ depth moves by $\leq 15\%$ across the certified window ($t = 0.0036$, 0.0040, t_m). The floor extent of the content (≥ 13 modes above 10^{-12} of the wall's driver amplitude) is $7.5 \cdot 10^{-3}$ / $8.1 \cdot 10^{-3}$ (512^2 / 1024^2) at $\sigma = 1.375$ and $1.08 \cdot 10^{-2}$ at both N at $\sigma = 0.875$ — the "thinner than $y = 0.01$" of the earlier draft is this extent; the amplitude e-folds twenty times closer to the wall. What is measured, exactly: the defect is largest at the wall row and decays through the content layer, the coarse grid's θ layer is thicker while its wall spectrum is steeper, and the ω layer is grid-converged; the mechanism stays unclaimed.

The defect's anisotropic signature. To separate the two resolutions the same runs were repeated on the anisotropic grids 512×1024 (coarse along the wall, fine across it) and 1024×512 (fine along, coarse across) at $\sigma = 1.375$ and the same driver damping, through a stage whose run loop is a mechanical copy of the certified loop with the two resolutions separated — first proved by reproducing the square 512^2 runs of the previous paragraph bit for bit on every stored key, at both σ. At $t_m = 0.004488$, on the wall row of the fixed grid with each row normalised by its own driver mode and all 26 modes of B^* usable, the difference of each grid from the square 1024^2 grid (the statistic of the previous paragraph; the square 512^2 grid reproduces its 0.572): the coarse-along / fine-across grid differs by 0.300 (mean +0.174), the fine-along / coarse-across grid by 0.084 (mean +0.049); the across-wall share $0.084/(0.300 + 0.084) = 0.219$ — the registered letter MIXED (bars 0.8 / 0.2 on the share; the labelled expectation, an across-wall effect from the seven Chebyshev rows inside the layer, did not obtain), one bar-step from the along-wall reading and the same under one common timing shift of $\pm 4.6 \cdot 10^{-5}$ (0.218 / 0.219 / 0.221; the mean-based share 0.216–0.220). The two shares sum to 0.38 of the square grid's 0.57 (means 0.22 of 0.34): about a third of the effect appears only when both directions are coarse. The control $\sigma = 0.875$ (read $9\,\mu$s earlier, at the end of the shortest of its four certified windows): 0.008 and 0.003 against the square grid's 0.019. The θ e-fold depth reads 6.0 / 5.6 / 5.4 / $5.2 \cdot 10^{-4}$ on the 512^2, 512×1024, 1024×512 and 1024^2 grids (on-grid, 7.5% steps — the coarse grid's excess splits one step per coarsened direction) and the ω depth $1.00 \cdot 10^{-3}$ on all four; each anisotropic grid's difference decays through the layer with the square grid's shape at its own amplitude ($0.30 \to 0.10$ and $0.084 \to 0.026$ from the wall to $y = 5 \cdot 10^{-3}$). What is measured, exactly: the 512^2 over-steepening of the wall temperature's spectrum is carried mostly by the along-wall resolution of the wall row, least by the across-wall resolution of the layer, and in part by both together. A labelled candidate, not claimed: products of B^* modes reach $k = 204$, above the 512-point dealiasing cutoff 170 and below the 1024-point cutoff 341. Read in the layer's own units — a re-reading of the previous paragraph's record, whose letter stands as fired — the 512^2 effect persists over 3.4 (512^2) to 3.9 (1024^2) e-fold depths (the floor extent is 12.4 depths) at 35–55% of its wall value by the mean (+0.34 at the wall, +0.23 at half a depth, +0.20 at three, +0.16 at four, +0.12 at the floor extent), while the max statistic spikes on six grid rows between $6 \cdot 10^{-4}$ and $1.1 \cdot 10^{-3}$ where individual modes pass through zero; the earlier letter's bars were set against the floor extent, an instrument quantity, which is why it read MIXED where the mean reads a layer-filling effect.

7 Where this sits in the literature

Statements below are cited for what the sources state, with their theorem numbers; a source not held at full text is cited at metadata level and marked "not read".

- **The dial and its critical exponent.** Hmidi, Keraani and Rousset [5, Theorem 1.1] prove for the Euler–Boussinesq system on $\mathbb{R}^2$ with the critical thermal diffusion $|D|\theta$ (their (1.2): $\partial_t v + v\cdot\nabla v + \nabla p = \theta e_2$, $\partial_t\theta + v\cdot\nabla\theta + |D|\theta = 0$, $\operatorname{div} v = 0$) that for $p \in]2,\infty[$, $v_0 \in B^1_{\infty,1}\cap W^{1,p}$ divergence-free and $\theta_0 \in B^0_{\infty,1}\cap L^p$ "there exists a unique global solution (v,θ) to the system (1.2)". Their introduction records that Hmidi and Zerguine [7] (not read; the statement as cited in [5, §1] and in [11, §1]) "proved the global well-posedness for fractional diffusion $D_\theta = -|D|^\alpha$ for $\alpha \in]1,2[$", the case $\alpha = 1$ being what [5] adds; their companion paper [6] treats velocity dissipation with an undiffused temperature, a different configuration from ours ([11, §1] cites the same result as "$\alpha = 0$, $\beta \in (1,2]$"). In the $|D|^\beta$ convention our σ is β: on $\mathbb{R}^2$ the dial is covered at $\sigma = 1$ by [5], for $1 < \sigma < 2$ by [7] as cited, and at $\sigma = 2$ — the Laplacian with inviscid velocity — by Chae [1] and Hou and Li [8], which [11, §1] records as "the global regularity to (1.1) with $\alpha = 2$, $\beta = 0$ or $\alpha = 0$, $\beta = 2$"; $\sigma < 1$ is not.

- **The regime map.** Stefanov, Wu, Xu and Ye [11, §1] state, for the 2D Boussinesq equations on $\mathbb{R}^2$ with $(-\Delta)^{\alpha/2}u$ and $(-\Delta)^{\beta/2}\theta$, the classification "the subcritical regime ($\alpha+\beta > 1$), the critical regime ($\alpha+\beta = 1$) and the supercritical regime ($\alpha+\beta < 1$)" and that "the global regularity problem for the supercritical regime remains largely out of reach, while substantial progress has been made in the critical and subcritical cases"; their Theorem 1.1 (global regularity in the subcritical regime) assumes $\alpha \in (0,1)$ and $\beta \in (0,1)$. Our configuration is $\alpha = 0$ (inviscid velocity): the rows $\sigma < 1$ of this bench are in the supercritical regime in their terms; the rows $\sigma \geq 1$ are covered on $\mathbb{R}^2$ by [5, 7, 1, 8].

- **Walls.** The bounded-domain result we hold is Zhao [13, Theorem 1.1]: for $\Omega \subset \mathbb{R}^2$ bounded and smooth, inviscid velocity with $U\cdot n|_{\partial\Omega} = 0$ and $\theta_t + U\cdot\nabla\theta = \kappa\Delta\theta$ with the DIRICHLET condition $\theta|_{\partial\Omega} = \bar\theta$ (his compatibility conditions (1.4)), H^3 data give "a unique solution (U,θ) to (1.1)–(1.2) globally in time", θ converging exponentially to $\bar\theta$ with constants exponentially large in $1/\kappa$ (his Remark 1.1: $\|\omega\|_\infty \leq \tilde c = O(\kappa^{-43}e^{21/\kappa})$). Our walls are no-flux for θ, our diffusion is fractional, and our $\sigma \geq 1$ rows sit at the critical exponent or below the Laplacian; the other boundary results we know of — Lai, Pan and Zhao [10] (not read; an initial-boundary-value problem for the viscous equations by its title) and Tan and Xu [12] (not read; the half plane with partial viscosity by its title) — and the whole-space partial-viscosity results [1, 8] treat velocity viscosity or the Laplacian. **We know of no published well-posedness result covering fractional thermal diffusion with no-flux walls at any σ; the $\sigma < 1$ rows are supercritical in the sense of [11] and carry a boundary.** For the Laplacian case $\sigma = 2$ with no-flux walls, Appendix A records a routine adaptation of [13]; it is an adaptation argument, not a new theorem. The measured laws of Section 2 ($\kappa_c(\sigma) = 0$ as an extrapolation; the $1/\kappa$ response) are consistent with and adjacent to the known Dirichlet-θ theory; the response law is exponentially sharper than the $e^{21/\kappa}$-class bound the theorem provides. No theorem is claimed here.

- **The corner.** The configuration is the 2D Boussinesq equations "on the upper half space" with the no-flow condition on the stream function, in which Chen and Hou [2, Theorem 1] prove that "there is a family of smooth initial data (θ_0,ω_0) with $\theta_0(x,y)$ even, $\omega_0(x,y)$ odd in x, $\theta_0(0,y) \equiv 0$, and finite-energy velocity u_0 such that the solution of the 2D Boussinesq equations develops a singularity in finite time $T < +\infty$", nearly self-similarly and stably in their symmetry class; our data are in that class on the strip, and our bench is the dissipative deformation of their setting. Chen, Huang and Li [3, §6] report numerically, in the same half-space setting, two-stage self-similar blowups with singular profiles from degenerate data. Both are inviscid; neither contains a dissipative statement, and nothing in this note bears on their theorems.

- **Non-uniqueness.** Chen and Yin [4, Theorem 2.1] prove, for the Boussinesq system on the torus with $(-\Delta)^\alpha$ on both equations, $1 \leq \alpha < (d+1)/2$, sharp non-uniqueness of weak solutions in $L^p_t L^\infty_x$ for $p < 2\alpha/(2\alpha - 1)$. A different object (weak solutions; dissipation on both equations; no walls); recorded as context.

- **Novelty grade.** The objects of Sections 2–6 are, to our knowledge, apparently uncomputed: a full-text census of the sources above (the terms fractional / dissipation / diffusion, critical / supercritical / subcritical, bounded domain / boundary / half / wall / Neumann / no-flux / Dirichlet, Hou / Luo / corner / hyperbolic, analytic / strip / spectrum / Fourier, saturation / threshold / blowup, numerics / simulation, every hit read in its sentence) found no statement about a dissipative response law at the Hou–Luo corner, a fractional-thermal-diffusion problem with walls, a σ-ordering of spectra under fractional dissipation, or a quantitative σ-dependence of a blowup-scenario observable; the configuration (fractional thermal diffusion at the Hou–Luo corner with walls) does not appear in the sources above. No "first" claim is made.

8 What is and is not claimed

Claimed (report-grade, on a certified bench): the σ-dial level and its certification; the saturated-branch collapse and the σ-extended saturation law with its domain bound; the monotone σ-ordering of the ω x-spectrum at matched time and band at five driver dampings, and of the θ-wall x-spectrum at the three where it clears the floor, N-converged on the ω spectrum to 0.15% mode by mode at resolution-class rows and to 0.9% at a saturated-class pair; the ordering's stability in time, the fixed-set backward growth of its magnitude (N-converged to 0.1%), and its weak decrease with the driver damping at a fixed time across the columns run (two overlapping four-column tables; the most-damped column's value at t_c N-converged to 0.08%); the withdrawal of the valley; the θ wall-layer resolution effect with its layer measured on one fixed physical grid at both resolutions (e-fold depth $\approx 5 \cdot 10^{-4}$, floor extent $\approx 8 \cdot 10^{-3}$; letter MIXED) and its anisotropic signature (with the wall-normal grid refined the effect keeps 0.30 of its 0.57, with the along-wall grid refined 0.08, a third needing both coarse; letter MIXED, leaning to the along-wall resolution). Not claimed: any analyticity-strip width; any functional form for the separation's time or X dependence; any mechanism for the θ-wall defect; any well-posedness statement; anything about the Navier–Stokes equations — nothing here is claimed to transfer to them. Open: the separation's X-dependence beyond $3\times$ and at later common times (no single common time exists at 512^2 for the least- and most-damped columns; a finer grid would move the least-damped column's certified window later); whether any law is warranted (only after more report rows, with registered priors); the θ layer's mechanism (its anisotropic signature is measured; the along-wall dealiasing cutoff is a labelled candidate, untested); the 1024^2 endgame migration of the θ maximum; the grid-convergence of the θ-wall separation at $\sigma \geq 1.375$ (a 2048^2 row, unpriced).

Data and code availability

The solver, the diagnostic records of every run reported here (including the runs behind the withdrawn valley), the registration files with their letters as fired, and the analysis scripts that emit every number in this note are archived at `https://github.com/UsernameLoad/clm-threshold-certificates` and deposited with a DOI at [**Zenodo DOI, inserted at posting**].

AI statement

This note was produced with an AI system. The mathematics, the computations and the text were generated by an AI research agent (Claude, Anthropic) working under my direction; I set the objectives and the verification rules, reviewed the results, and take full responsibility for the content. Every number in this note is reproducible from the scripts and records archived with the code. No AI system is an author of this note.

A The no-flux adaptation of Zhao's theorem at $\sigma = 2$

This appendix records, at argument level, that the $\sigma = 2$ configuration of this bench — inviscid velocity, $\theta_t + U \cdot \nabla\theta = \kappa\Delta\theta$ with no-flux walls — is covered by a routine adaptation of [13, Theorem 1.1], whose boundary condition is Dirichlet. It settles for our own purposes the question "is the no-flux configuration known-regular at $\sigma = 2$" at the level of a written adaptation; a submission-grade version would write out the five substituted estimates in full, which is mechanical and contains no new idea. It does not touch $\sigma < 2$.

> **Lemma.** Let Ω be either a smooth bounded domain in $\mathbb{R}^2$ or the periodic strip $\mathbb{T}_L \times [0,1]$. Consider 2D inviscid heat-conductive Boussinesq flow $U_t + U \cdot \nabla U + \nabla P = \theta e_2$, $\nabla \cdot U = 0$, $\theta_t + U \cdot \nabla\theta = \kappa\Delta\theta$ ($\kappa > 0$), with $U \cdot n = 0$ and the no-flux condition $\partial\theta/\partial n = 0$ on $\partial\Omega$ (data smooth, divergence-free, satisfying the Neumann-parabolic compatibility conditions). Then the solution is global and smooth; $\theta \to \bar{\theta} := |\Omega|^{-1} \int \theta_0$ exponentially in H^3; and $\|\omega(\cdot,t)\|_\infty \le \tilde{c}(\kappa)$ uniformly in t, with the same κ-dependence class as the Dirichlet case.

Adaptation, step by step against [13, §3].

1. *The conserved mean replaces the Dirichlet zero.* $\frac{d}{dt} \int \theta = \oint \kappa\, \partial\theta/\partial n - \oint (U \cdot n)\theta = 0$ under exactly our two boundary conditions, so $\bar{\theta}$ is a constant of motion; set $\varphi := \theta - \bar{\theta}$. Then φ solves the same equation ($\bar{\theta}$ is constant in space and time) and the buoyancy sees only φ: $\theta_x \equiv \varphi_x$, and $\theta e_2 = \varphi e_2 + \bar{\theta} e_2$ with $\bar{\theta} e_2$ a gradient, absorbed into P. Every estimate below runs on φ.

2. *Poincaré becomes Poincaré–Wirtinger (his (3.2)).* $\|\varphi\| \le c_0' \|\nabla\varphi\|$ for mean-zero φ, classical on bounded smooth Ω and on the strip. His Lemma 3.1 then gives $\|\varphi(t)\|^2 \le \|\varphi_0\|^2 e^{-2\beta_0 t}$ verbatim: in (3.1) the transport term dies by $\nabla \cdot U = 0$ with $\oint (U \cdot n)\theta^2 = 0$ (shared boundary condition), and the dissipation boundary term $\oint (\partial\theta/\partial n)\theta$ dies under no-flux exactly where Dirichlet killed it.

3. *Dirichlet elliptic regularity becomes Neumann elliptic regularity (his Lemma 2.2, used at (3.14) and its H^2/H^3 repeats).* $\kappa\Delta\varphi = f := \varphi_t + U \cdot \nabla\varphi$ with $\partial\varphi/\partial n = 0$; the compatibility $\int f = \frac{d}{dt} \int \varphi + \int U \cdot \nabla\varphi = 0$ holds automatically (step 1), and $\|\varphi\|_{W^{m+2,p}} \le (c_5'/\kappa)\|f\|_{W^{m,p}}$ for mean-zero φ is the classical Neumann estimate (Agmon–Douglis–Nirenberg; Grisvard) on smooth Ω; on the strip, Fourier in x and the one-dimensional Neumann problem give it directly (no corners — the strip case is the easier one).

4. *Boundary-term symmetry at every differentiated level (his (3.41)–(3.42) and the Remark 3.1 repeats).* His only boundary inputs are $\{\theta|_\partial = 0 \Rightarrow \theta_t|_\partial = 0$ on the fixed boundary$\}$ and $\{U \cdot n = 0 \Rightarrow U_t \cdot n = 0\}$. The Neumann twins hold by the same time differentiation on the fixed boundary: $\partial\theta/\partial n = 0 \Rightarrow \partial\theta_t/\partial n = 0$; $U_t \cdot n = 0$ is shared. Every boundary integral in the $H^1/H^2/H^3$ chain (tested with φ_t, $\Delta\varphi$-type quantities, and the time-differentiated equation) dies under the Neumann twins exactly where the Dirichlet originals died. Checked term by term through his Lemmas 3.1–3.5: no estimate uses the sign or the trace value of θ on $\partial\Omega$, only that the boundary integrals vanish.

5. *The velocity chain is untouched.* His Lemmas 2.3 (div-curl with $U \cdot n = 0$), 2.5 (Euler local theory), 3.2 and 3.4 ((3.31)–(3.32): $\omega_t + U \cdot \nabla\omega = \varphi_x$ tested with $|\omega|^{p-2}\omega$ — no θ-boundary contact) are velocity-side and identical; on the strip, Lemma 2.3 by Fourier in x. The

weak-existence fixed point (his Lemma 2.6) runs on the Neumann heat semigroup with mean-zero spaces ($H^1 \cap \{\text{mean } 0\}$ replacing H^1_0) — standard.

Conclusion: Zhao's Theorem 3.1 chain closes with constants c_0', c_5' in place of c_0, c_5; the decay target is $\bar{\theta}$ instead of the boundary value; the κ-dependence classes (his $O(\kappa^{-1}e^{1/\kappa})$-type ladders) are unchanged in form. The compatibility-condition line in the data hypothesis is the standard Neumann-parabolic one, replacing his (1.4). Consequence for the claim language of Section 7: the no-flux configuration at $\sigma = 2$ is argued-regular, not mystery-open, and the measured κ-response law of Section 2 is quantitative structure inside a regime that this argument covers, exponentially sharper than the provable $e^{21/\kappa}$-class vorticity bound.

References

[1] D. Chae, *Global regularity for the 2D Boussinesq equations with partial viscosity terms*, Adv. Math. 203 (2006) 497–513, DOI 10.1016/j.aim.2005.05.001.

[2] J. Chen, T. Y. Hou, *Stable nearly self-similar blowup of the 2D Boussinesq and 3D Euler equations with smooth data I: Analysis*, arXiv:2210.07191 (v4, 2026). (Part II, the rigorous numerics, is Multiscale Model. Simul. 23 (2025) 25–130, DOI 10.1137/23M1580395 — metadata only, not read.)

[3] B. Chen, D. Huang, X. Li, *Novel self-similar finite-time blowups with singular profiles of the 1D Hou–Luo model and the 2D Boussinesq equations: a numerical investigation*, arXiv:2604.01868 (2026).

[4] Z. Chen, Z. Yin, *Sharp non-uniqueness for the Boussinesq equation with fractional dissipation*, arXiv:2511.22023 (2025).

[5] T. Hmidi, S. Keraani, F. Rousset, *Global well-posedness for Euler–Boussinesq system with critical dissipation*, Comm. Partial Differential Equations 36 (2011) 420–445, DOI 10.1080/03605302.2010.518657; arXiv:0903.3747.

[6] T. Hmidi, S. Keraani, F. Rousset, *Global well-posedness for a Boussinesq–Navier–Stokes system with critical dissipation*, J. Differential Equations 249 (2010) 2147–2174, DOI 10.1016/j.jde.2010.07.008. (The companion paper: velocity dissipation, θ undiffused.)

[7] T. Hmidi, M. Zerguine, *On the global well-posedness of the Euler–Boussinesq system with fractional dissipation*, Physica D 239 (2010) 1387–1401, DOI 10.1016/j.physd.2009.12.009. (Not read; cited as stated in [5, §1] and [11, §1].)

[8] T. Y. Hou, C. Li, *Global well-posedness of the viscous Boussinesq equations*, Discrete Contin. Dyn. Syst. 12 (2005) 1–12, DOI 10.3934/dcds.2005.12.1.

[9] G. Luo, T. Y. Hou, *Toward the finite-time blowup of the 3D axisymmetric Euler equations: a numerical investigation*, Multiscale Model. Simul. 12 (2014) 1722–1776, DOI 10.1137/140966411. (The driver data and the strip period.)

[10] M.-J. Lai, R. Pan, K. Zhao, *Initial boundary value problem for two-dimensional viscous Boussinesq equations*, Arch. Ration. Mech. Anal. 199 (2011) 739–760, DOI 10.1007/s00205-010-0357-z. (Not read.)

[11] A. Stefanov, J. Wu, X. Xu, Z. Ye, *Global regularity of the 2D fractional Boussinesq equations with subcritical dissipation*, arXiv:2606.03680 (2026).

[12] Z. Tan, X. Xu, *Global regularity for 2D Boussinesq equations with partial viscosity in the half plane*, Math. Methods Appl. Sci. 46 (2023) 6484–6505, DOI 10.1002/mma.8920. (Not read; title only.)

[13] K. Zhao, *2D inviscid heat conductive Boussinesq equations on a bounded domain*, Michigan Math. J. 59 (2010) 329–352, DOI 10.1307/mmj/1281531460.